Abbas Taqvi

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Professional Summary

- Data engineer and M.S. Data Science student at Milano-Bicocca seeking a machine learning internship in Italy. Built a LangGraph Text2SQL system that reduced analyst turnaround by 41% and a PyTorch normalizing-flow framework that reduced parameter inference from roughly 300 minutes to about 2 seconds. Brings experience building cloud data pipelines, training neural models and connecting AI systems to business data.

Education

- **Università degli Studi di Milano-Bicocca** — Milan, Italy
  Master of Science in Data Science — *Oct 2025 – Present*
- **Habib University** — Karachi, Pakistan
  Bachelor of Science in Computer Science; GPA: 3.31; 100% merit scholarship (HU-TOPS) — *Aug 2020 – May 2024*

Experience

- **Logiciel Services LLC** — United States (Remote)
  Data Engineer I — *May 2024 – Nov 2025*
  - **Text2SQL System**: Built Metosi, a multi-agent chatbot in Python, LangChain and LangGraph, connecting natural-language requests to Trino SQL queries through a REST API; reduced analyst turnaround by 41%.
  - **Data Platform**: Built a Kafka, Spark (Scala), S3, Trino and Airflow platform for financial data; simplified 12 components to 5 and enabled sub-second queries over 30B+ records with appropriate filters.
  - **ETL Performance**: Developed Ferroflow, a Rust framework with reusable processors and composable pipelines, reducing daily FIX-log processing from 500 to 50 minutes (10× faster).
- **Pullflow Inc.** — San Jose, CA (Remote)
  DevOps Intern — *Jul 2023 – Aug 2023*
  - **Kubernetes Migration**: Migrated production infrastructure from AWS EKS to MicroK8s using Docker, Helm and Traefik; deployed a self-hosted analytics dashboard with a zero-downtime cutover.
- **United Bank Limited** — Karachi, Pakistan
  Data Engineering Intern — *May 2023 – Jun 2023*
  - **Migration Pipeline**: Built a cross-region Informatica pipeline into a Hive data lake; validated production AI pipelines against regional schemas across 3 countries.
- **ImproData** — Pinedale, WY (Remote)
  Junior Data Engineer — *Jan 2021 – May 2022*
  - **Data Pipelines**: Built pipelines on Azure Databricks (PySpark) and automated ingestion via GCP Cloud Functions (JSON → GCS → BigQuery), eliminating 3 hours of daily manual work.

Projects

- **Astroflow: Bayesian Inference with Normalizing Flows**: Built a PyTorch framework for binary black-hole population modeling; designed neural summarizers and feature extractors, optimized training loops, and reduced parameter inference from roughly 300 minutes to about 2 seconds. Built a Django interface for experiments and posterior visualization.
- **Explainable AI with Diffusion Models**: Explored explainability techniques using diffusion models in a computer vision course at Habib University.

Skills

- **Programming**: Python, SQL, C/C++ (CUDA), Rust, Scala
- **Machine Learning**: PyTorch, TensorFlow, JAX, NumPyro, normalizing flows, Hugging Face transformers
- **AI Systems**: LangChain, LangGraph, RAG, Text2SQL, multi-agent systems
- **Data & Deployment**: Pandas, NumPy, Spark, Airflow, Trino, AWS, Docker, Kubernetes, Git
- **Spoken Languages**: English (C1), Italian (A2)

Achievements & Test Scores

- **Research Poster**: Astroflow, The Computing Conference 2025 (SAI Conferences), London, UK.
- **Test Scores**: IELTS 8.0/9.0; GRE Quantitative 165/170
- **Competitive Programming**: ICPC Asia Topi Regionals 2022: 19th place; Google Kickstart (May 2022): 6th place country-wide

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